Neonatal Inflammatory Skin and Bowel Disease, Type 1 (NISBD1) — Comprehensive Disease Report

Disease: Inflammatory Skin and Bowel Disease, Neonatal, 1 (NISBD1) **MONDO ID:** MONDO:0013693 · **OMIM:** #614328 **Causal gene: ADAM17** (a disintegrin and metalloproteinase 17; also called **TACE**, TNF-α-converting enzyme; HGNC:195) **Inheritance:** Autosomal recessive · **Category:** Mendelian

Summary

Neonatal Inflammatory Skin and Bowel Disease type 1 (NISBD1) is an ultra-rare, autosomal-recessive Mendelian disorder caused by **biallelic loss-of-function (LOF) variants in ADAM17** (TACE). ADAM17 is the principal cell-surface "sheddase" that proteolytically releases the active, soluble forms of **EGFR ligands** (TGF-α, amphiregulin, HB-EGF) and of **TNF-α** from the plasma membrane. When ADAM17 activity is lost, epithelial tissues fail to receive EGFR-ligand–driven developmental and barrier-maintenance signals, and inflammatory cytokine signaling becomes dysregulated. The result is a neonatal-onset multi-system disease dominated by inflammatory skin disease, inflammatory/secretory bowel disease, and hair/nail abnormalities.

Clinically, affected neonates present with a **collodion membrane at birth, diffuse scaly erythematous skin / erythroderma, evanescent pustules, chronic watery diarrhea, failure to thrive, and recurrent bacterial infections/sepsis**. Additional features include **fragile or absent hair (atrichia), nail dystrophy, esophageal strictures**, and variable multi-organ involvement (hepatitis, cardiomyopathy, hypertension). Mortality in infancy and childhood is substantial, driven by sepsis and cardiac disease. Fewer than ~10 families have been reported worldwide since the disease was first defined in 2011, and consanguinity is a recurrent feature.

The mechanistic basis is now well-supported. ADAM17-null mice phenocopy EGFR/TGF-α-pathway deficiency; conditional deletions reproduce the hair-follicle, skin-barrier, and colonic-inflammation phenotypes; and the obligate ADAM17 cofactors iRhom1/iRhom2 (RHBDF1/

RHBDF2) tune enzyme maturation and activity — with *gain-of-function* iRHOM2 mutations causing the mirror-image skin disease tylosis with esophageal cancer. Crucially, **NISBD1 (ADAM17 loss) and NISBD2 (EGFR loss) form an allelic series along a single pathway**, confirming that the pathology runs through the **ADAM17 → EGFR-ligand → EGFR** epithelial axis. Management is supportive plus anti-inflammatory/biologic therapy (notably combined anti-TNF-α + IL-12/23 blockade for skin, budesonide for gut); no curative therapy exists.

> **Note on disease identity.** The identifiers in the task (MONDO:0013693 + ADAM17) point unambiguously to NISBD1. It must be distinguished from the phenotypically overlapping **NISBD2 (OMIM #616069)**, caused by biallelic variants in **EGFR** — a different gene acting immediately **downstream** of ADAM17 in the same pathway. That very overlap is strong evidence that NISBD1 pathology is EGFR-pathway driven. All content below refers to the ADAM17 (NISBD1) entity.

Key Findings

Finding 1 — NISBD1 is caused by biallelic loss-of-function variants in ADAM17 (TACE)

The founding report by **Blaydon et al. (NEJM 2011)** mapped autosomal-recessive neonatal inflammatory skin and bowel lesions in a consanguineous sibpair to a loss-of-function mutation in ADAM17. Patient peripheral-blood mononuclear cells (PBMCs) showed **impaired LPS-induced TNF-α release but elevated IL-1β and IL-6**, consistent with loss of the enzyme that normally sheds membrane pro-TNF-α. The authors concluded: *"We identified a loss-of-function mutation in ADAM17, which encodes a disintegrin and metalloproteinase 17 (also called tumor necrosis factor α [TNF-α]-converting enzyme, or TACE), as the probable cause of this syndrome"* (PMID: 22010916).

The gene–disease relationship has been reproduced in multiple independent families with diverse LOF alleles: - **Bandsma et al. (2015)**: homozygous frameshift `NM_003183.4:c.308dupA`, producing a premature stop codon — *"We identified a novel homozygous frameshift mutation in ADAM17 (NM_003183.4:c.308dupA) leading to a premature stop codon"* (PMID: 25804906). - **Imoto et al. (2021)**: functionally confirmed compound-heterozygous **missense** variants `c.1699T>C (p.Cys567Arg)` and `c.1799G>A (p.Cys600Tyr)`, validated in an Adam10/17 double-

knockout mouse-fibroblast shedding assay (PMID: 33953303). - **Samuelov et al. (2022)**: a complex compound-heterozygous defect — a large deletion of exons 6–7 plus a splice variant causing exon-17 skipping (PMID: 34993966). - **Malhi et al. (2026)** and **Chang et al. (2026)**: additional homozygous alleles, including `c.436G>A; p.Gln146Lys`, expanding the phenotype (PMID: 42015567, PMID: 40968583).

Evidence type: human clinical + in vitro functional confirmation. The convergence of nonsense, frameshift, missense, large-deletion, and splice alleles — all LOF — firmly establishes ADAM17 haploinsufficiency/null as the cause.

Finding 2 — ADAM17 is the major sheddase of EGFR ligands and TNF-α; its loss impairs epithelial development

ADAM17 processes **more than 80 substrates**, most importantly the EGFR-family growth factors and pro-TNF-α (Imoto 2021, PMID: 33953303). **Peschon et al. (Science 1998)** showed that TACE/ADAM17-null mice fail to process TGF-α and phenocopy EGFR-pathway deficiency, concluding: *"The phenotype of mice lacking TACE suggests an essential role for soluble TGFalpha in normal development and emphasizes the importance of protein ectodomain shedding in vivo"* (PMID: 9812885). **Edwards et al. (2008)** summarize the enzyme's physiological role: *"ADAM-17 is required for generation of the active forms of Epidermal Growth Factor Receptor (EGFR) ligands, and its function is essential for the development of epithelial tissues"* (PMID: 18762209).

This establishes the mechanistic "why": ADAM17 loss simultaneously (a) **removes the proteolytic step that generates active EGFR ligands**, starving epithelial tissue of EGFR-driven proliferation/differentiation/barrier signals, and (b) **alters TNF-α and other cytokine processing**, dysregulating inflammation.

Finding 3 — iRhom2/iRhom1 (RHBDF2/RHBDF1) are obligate cofactors controlling ADAM17 maturation and activation

ADAM17 does not act alone. Cryo-EM structures demonstrate that **iRhom2 promotes ADAM17 trafficking and enzyme maturation and retains the inhibitory prodomain to prevent premature activation**. **Lu et al. (2024)**: *"iRhom2 interacts with the ADAM17 TMD to promote ADAM17 trafficking and enzyme maturation"* (PMID: 38781971); **Maciag et al. (2025)** add structural detail on prodomain-mediated inhibition and a cytoplasmic re-entry loop involved in activation (PMID: 40512800).

The clinical mirror image is instructive: **gain-of-function iRHOM2 mutations cause tylosis with esophageal cancer (TOC)** by *increasing* ADAM17 maturation/activity. **Brooke et al. (2014)**: *"TOC-associated mutations in iRHOM2 cause an increase in the maturation and activity of ADAM17 in epidermal keratinocytes, resulting in significantly upregulated shedding of ADAM17 substrates, including EGF-family growth factors and pro-inflammatory cytokines"* (PMID: 24643277). NISBD1 (too little ADAM17 activity) and TOC (too much) thus bracket the same axis — strong orthogonal support for the pathway.

Finding 4 — NISBD1 skin inflammation responds to combined anti-TNF-α and IL-12/23 blockade

Samuelov et al. (2022) reported a NISBD1 patient with erythroderma, atrichia, nail dystrophy, esophageal strictures, intractable diarrhea and failure to thrive (complex compound-heterozygous ADAM17 defect). *"Skin manifestations dramatically improved in response to combined anti-tumour necrosis factor-α and interleukin-12/23 blockade, while gastrointestinal symptoms were controlled with budesonide"* (PMID: 34993966). This provides the best-documented targeted therapeutic strategy and reinforces the inflammatory cytokine component of the disease.

Finding 5 — Mouse Adam17 models recapitulate key NISBD1 features via EGFR-ligand shedding

Model	Phenotype	NISBD1 feature modeled	PMID
Adam17-null (germline)	Perinatal/embryonic lethal; open-eye-at-birth; skin/epithelial defects; phenocopies EGFR/TGF-α deficiency	Skin barrier / epithelial immaturity	9812885
Tace/Sox9 conditional (hair-follicle stem cells)	Brittle hair, prolonged anagen, progressive whole-body hair loss, loss of CD34+ bulge cells	Fragile hair / atrichia	22696231
Systemic/ intestinal Adam17 deletion	Severe DSS colitis; impaired EGFR activation, epithelial proliferation, mucus/goblet-cell production, barrier function	Bowel inflammation	27077118
iRhom gain/loss (cub/toc)	Skin/hair phenotypes dependent on amphiregulin + Adam17	ADAM17→AREG→EGFR dependence	37282854

Nagao et al. (2012): *"Tace/Sox9 mice were born with brittle hair with prolonged anagen phase. They underwent diffuse, progressive, and ultimately whole-body hair loss by 20 weeks old"* (PMID: 22696231). **Shimoda et al. (2016)**: *"Mice with systemic deletion of Adam17 developed severe dextran sulfate sodium-induced colitis"* (PMID: 27077118). These models causally link ADAM17 loss to the skin, hair, and gut arms of the human disease through impaired EGFR signaling.

Finding 6 — Phenotype spectrum: neonatal erythroderma/pustules, intractable diarrhea, failure to thrive, infections, hair/nail defects, multi-organ involvement

The core triad across families is **chronic watery diarrhea, failure to thrive, and skin disease**. **Chang et al. (2026)**: *"characterized by chronic diarrhea, failure to thrive, rash, and recurrent bacterial infections. Cutaneous manifestations include the presence of a collodion membrane at birth, diffuse scaly erythematous patches and thin plaques, evanescent pustules, and/or erythroderma"* (PMID: 40968583). Additional features include fragile/absent hair (atrichia), nail dystrophy, and esophageal strictures (Samuelov 2022, PMID: 34993966).

Multi-organ dysfunction is documented. **Bandsma et al. (2015)**: *"the patient developed unexplained systolic hypertension and nonspecific hepatitis with apoptosis"*, with recurrent sepsis causing death at 10 months and diminished T-cell TNF-α/IL-2 production (PMID: 25804906). Cardiac disease is a recognized cause of death (see Finding 7). An atypical patient presented with pyoderma-gangrenosum-like ulcerations and pyostomatitis-like mucosal involvement but **no bowel disease** (PMID: 42015567), underscoring variable expressivity.

Finding 7 — Ultra-rare autosomal-recessive disorder with complete penetrance, variable expressivity, and high early mortality

Inheritance is autosomal recessive (OMIM #614328), requiring biallelic ADAM17 LOF; **fewer than ~10 families have been reported worldwide since 2011**, frequently consanguineous. Penetrance appears complete but **expressivity varies from survivable chronic disease to fatal infantile course**. Mortality drivers include sepsis (death at 10 months — Bandsma 2015) and cardiac disease. **Blaydon et al. (2011)**: *"The girl died suddenly at 12 years of age from parvovirus B19-associated myocarditis; her brother had mild cardiomyopathy"* (PMID: 22010916). Diagnosis is genomic — WES/WGS or targeted ADAM17 sequencing plus deletion/splicing analysis; **a 2024 study** supports an upfront WGS approach for these barrier/immune disorders: *"Upfront genome-wide analysis by whole genome sequencing (WGS) will shorten the time to diagnosis, save patients from unnecessary investigations, and reduce morbidity and mortality"* (PMID: 39381601).

Finding 8 — NISBD1 (ADAM17) and NISBD2 (EGFR) are an allelic series along one pathway

NISBD2 (OMIM #616069) is an autosomal-recessive, EGFR-associated early-infantile disease. **Karabinos et al. (2025)**: *"Neonatal inflammatory skin and bowel disease 2 is an epidermal growth factor receptor (EGFR)-associated autosomal recessive early-infantile disease with inflammatory skin, alopecia, progeroid features, electrolyte imbalance, recurrent infections, and premature death"* (PMID: 40040597). EGFR-null mice show postnatal lethality with epithelial immaturity and intestinal lesions: *"The mildest form of EGF-R inactivation leads to epithelial immaturity and postnatal death due to respiratory failure and necrotizing enterocolitis-like lesions in the intestine"* (PMID: 9562519). Because ADAM17 generates the soluble ligands that activate EGFR, the near-identical phenotypes place NISBD1 and NISBD2 at **consecutive steps of a single ADAM17 → EGFR-ligand → EGFR epithelial axis** — the strongest internal validation of the proposed mechanism.

Section-by-Section Report

1. Disease Information

NISBD1 is a Mendelian, autosomal-recessive genodermatosis-plus-enteropathy of neonatal onset, caused by loss of ADAM17 sheddase function. It combines severe inflammatory skin disease, inflammatory/secretory bowel disease, hair and nail defects, and heightened infection susceptibility.

Key identifiers: - **MONDO:** MONDO:0013693 - **OMIM:** #614328 (Inflammatory skin and bowel disease, neonatal, 1) - **Orphanet:** NISBD / ADAM17 deficiency (listed under neonatal inflammatory skin and bowel disease) - **ICD-10:** no specific code; mapped under congenital ichthyosiform erythroderma + inflammatory enteropathy categories (e.g., Q80.x / K52.x as closest approximations) - **MeSH:** no dedicated descriptor; indexed via ADAM17 Protein, erythroderma, and inflammatory bowel disease terms

Synonyms / alternative names: NISBD1; ADAM17 deficiency; neonatal inflammatory skin and bowel disease due to ADAM17 mutation; inflammatory skin and bowel disease, neonatal, type 1.

Information source: Disease-level knowledge derives from aggregated OMIM/Orphanet entries and a small number of **individual-patient case reports/series** (fewer than ~10 families). There is no large EHR-derived cohort; all data are case-based.

2. Etiology

Primary cause: Genetic — biallelic (homozygous or compound-heterozygous) loss-of-function variants in **ADAM17** (Findings 1–2). The disease is fully genetic; no environmental or infectious cause initiates it, although infections are frequent *complications*.

Genetic risk factors: The causal variants themselves (LOF ADAM17 alleles) are the entire genetic risk. **Consanguinity** markedly raises risk by increasing homozygosity for rare recessive LOF alleles. No common susceptibility loci or GWAS signals apply (this is monogenic).

Modifier genes: iRhom1/iRhom2 (RHBDF1/RHBDF2) and downstream EGFR-pathway genes are biologically plausible modifiers of residual ADAM17 activity/expressivity (Finding 3), but no formal human modifier has been proven. Variable expressivity (e.g., absence of bowel disease in one patient, PMID: 42015567) suggests modifiers and/or allele-specific residual function.

Environmental risk factors: None causal. Postnatal microbial exposure and barrier breakdown drive the infectious complications. **Protective factors:** none genetically established; supportive measures (infection prophylaxis, nutrition) mitigate severity. **Gene–environment interactions:** the genetic barrier/immune defect sensitizes patients to environmental microbes (bacterial skin infection, sepsis) — a gene→environment susceptibility relationship rather than a modifying GxE effect.

3. Phenotypes

Phenotype	Type	Onset	Frequency	Suggested HPO
Erythroderma / scaly erythematous patches	Clinical sign (skin)	Neonatal	Very frequent	HP:0001019 (Erythroderma)
Collodion membrane at birth	Physical manifestation	Congenital	Frequent	HP:0007479 / collodion membrane
Pustular skin lesions (evanescent)	Clinical sign	Neonatal	Frequent	HP:0200039 (Pustule)
Chronic watery diarrhea	Symptom (GI)	Neonatal	Very frequent	HP:0002028 (Chronic diarrhea)
Failure to thrive	Clinical sign	Neonatal/ infancy	Very frequent	HP:0001508 (Failure to thrive)
Recurrent bacterial infections / sepsis	Clinical sign	Infancy	Frequent	HP:0002718 (Recurrent bacterial infections)
Fragile hair / atrichia	Physical manifestation	Infancy	Frequent	HP:0011362 / HP:0001596 (Alopecia)
Nail dystrophy (paronychia)	Physical manifestation	Infancy	Frequent	HP:0008404 (Nail dysplasia)
Esophageal strictures	Clinical sign	Variable	Occasional	Esophageal stricture
Hepatitis with apoptosis	Lab/pathology	Variable	Occasional	HP:0200119 (Hepatitis)
Cardiomyopathy / myocarditis	Clinical sign	Childhood	Occasional (fatal)	HP:0001638 (Cardiomyopathy)
Systolic hypertension	Lab/sign	Variable	Rare	HP:0004421 (Systolic hypertension)

Severity: Severe, often life-threatening in infancy; **progression:** chronic with episodic flares; **variable expressivity** documented. **Quality-of-life impact** is profound in affected infants (chronic skin pain/pruritus, malabsorption, hospitalization, infection burden), though no formal QoL instrument data exist for this ultra-rare disease.

4. Genetic / Molecular Information

- **Causal gene:** ADAM17 (chromosome 2p25.1; OMIM *603639; HGNC:195; UniProt P78536).
- **Variant classification:** Reported variants are **pathogenic/likely pathogenic** (ACMG/AMP), functionally confirmed in several cases.
- **Variant types reported:** nonsense/frameshift (`c.308dupA`), missense (`p.Cys567Arg`, `p.Cys600Tyr`, `p.Gln146Lys`), large multi-exon deletion (exons 6–7), and splice variant (exon-17 skipping). The cysteine-substitution missense variants disrupt disulfide-dependent folding of the metalloprotease/disintegrin region, abolishing shedding (Imoto 2021).
- **Allele frequency:** All causal alleles are ultra-rare/private in gnomAD; biallelic LOF of ADAM17 is essentially absent in the general population, consistent with severe recessive disease.
- **Origin:** Germline, recessive. No somatic-mosaicism cases reported.
- **Functional consequence: Loss of function** (reduced/absent ADAM17 sheddase activity). Complete LOF in originally described patients; missense alleles cause functional LOF.
- **Modifier genes / epigenetics / chromosomal abnormalities:** No epigenetic mechanism or chromosomal abnormality is implicated; iRhom cofactors are mechanistic modulators (Finding 3) but not reported disease modifiers in patients.

5. Environmental Information

NISBD1 has **no environmental or infectious cause**. Environmental factors are relevant only as **triggers of complications**: cutaneous/systemic bacterial infection and sepsis arise because of the defective skin barrier and dysregulated immunity. No toxin, occupational exposure, lifestyle, or dietary factor is etiologic. Infectious agents (e.g., *Staphylococcus* and other skin/gut flora; parvovirus B19 in one fatal myocarditis case) act opportunistically on the genetically impaired host.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic LOF variants in ADAM17** (nonsense/frameshift/missense/deletion/splice) reduce or abolish ADAM17 (TACE) protein/activity. *(demonstrated — Findings 1–2)*
2. Loss of ADAM17 **results in failure of ectodomain shedding** of membrane-anchored EGFR ligands (TGF-α, amphiregulin, HB-EGF) and of pro-TNF-α. *(demonstrated — Peschon 1998; Edwards 2008)*

3. Failure to release soluble EGFR ligands **leads to deficient EGFR (ErbB1) signaling** in epithelial tissues. *(demonstrated in models — Nagao 2012; Shimoda 2016)*
4. Deficient EGFR signaling **results in impaired epithelial proliferation, differentiation, barrier formation, and goblet-cell/mucus production**. *(demonstrated)*
5. **Branch A (skin/hair):** → defective stratum-corneum barrier and hair-follicle stem-cell niche → **collodion membrane, erythroderma, pustules, fragile hair/atrichia, nail dystrophy**.
6. **Branch B (gut):** → impaired colonic epithelial protection → **intestinal inflammation, chronic secretory/inflammatory diarrhea, failure to thrive**.
7. Barrier breakdown plus dysregulated cytokine processing (loss of orderly TNF-α shedding; elevated IL-1β/IL-6; impaired T-cell TNF-α/IL-2) **leads to recurrent bacterial infection and sepsis**. *(demonstrated — Blaydon 2011; Bandsma 2015)*
8. Systemic inflammation and still-incompletely-defined mechanisms **contribute to multi-organ involvement** — hepatitis, hypertension, cardiomyopathy/myocarditis. *(observed; mechanism partly inferred)*
9. Sepsis and cardiac disease **are the principal causes of early mortality**. *(demonstrated)*

Molecular pathways: EGFR/ErbB (MAPK-ERK, PI3K-AKT downstream); TNF/TNF-receptor signaling. **Cellular processes:** epithelial proliferation/differentiation (GO:0030855 epithelial cell differentiation), keratinocyte differentiation (GO:0030216), inflammatory response (GO:0006954), hair-follicle development (GO:0001942), membrane-protein ectodomain proteolysis (GO:0006509). **Protein dysfunction:** loss of metalloprotease catalytic function (GO:0004222 metalloendopeptidase activity) due to truncation, misfolding (cysteine variants), or absent protein. **Immune involvement:** combined barrier-immunodeficiency and dysregulated innate cytokine milieu (elevated IL-1β/IL-6, impaired TNF-α release). **Tissue damage:** inflammatory epithelial injury of skin and gut from loss of trophic/protective EGFR signaling plus secondary infection.

Upstream vs downstream: ADAM17 LOF (upstream) → EGFR-ligand deficiency → EGFR signaling loss → epithelial failure and inflammation (downstream). The NISBD1↔NISBD2 allelic series (Finding 8) confirms ADAM17 is immediately upstream of EGFR.

Suggested CL cell types: keratinocyte (CL:0000312), basal epidermal cell (CL:0002187), hair-follicle stem/bulge cell, intestinal epithelial cell (CL:0002563), goblet cell (CL:0000160), T cell (CL:0000084), macrophage (CL:0000235).

7. Anatomical Structures Affected

- **Primary organs: Skin** (UBERON:0002097) and **intestine/colon** (UBERON:0000160 / UBERON:0001155); **hair follicle** (UBERON:0002073); **nail** (UBERON:0001705); **esophagus** (UBERON:0001043).
- **Secondary / systemic:** liver (UBERON:0002107 — hepatitis), heart (UBERON:0000948 — cardiomyopathy/myocarditis), vasculature (hypertension), immune system.
- **Body systems:** integumentary, digestive, cardiovascular, hepatic, immune.
- **Tissue/cell level:** stratified squamous epithelium of skin/esophagus; intestinal columnar epithelium with goblet cells; hair-follicle stem-cell (bulge, CD34+) compartment.
- **Subcellular:** plasma membrane (GO:0005886) — the site of ADAM17 sheddase activity; secretory/trafficking compartments (ER/Golgi) relevant to ADAM17 maturation via iRhom cofactors.
- **Lateralization:** bilateral/generalized skin and diffuse gut involvement (not lateralized).

8. Temporal Development

- **Onset: Neonatal / congenital** (collodion membrane at birth; skin and GI disease within the first days–weeks of life).
- **Onset pattern:** acute neonatal presentation, chronic course.
- **Progression:** chronic, with episodic inflammatory flares and recurrent infections; hair/nail and GI features persist.
- **Disease course:** chronic lifelong in survivors; high early mortality (infancy–childhood). No spontaneous remission; partial, treatment-induced improvement reported with biologics (Finding 4).
- **Critical periods:** the neonatal–infancy window is both the period of greatest vulnerability (sepsis, failure to thrive) and the key window for supportive/anti-inflammatory intervention.

9. Inheritance and Population

- **Inheritance:** Autosomal recessive (OMIM #614328).
- **Penetrance:** complete (biallelic LOF → disease). **Expressivity:** variable (survivable chronic disease to fatal infantile course; one patient lacked bowel disease).
- **Epidemiology:** ultra-rare; **<10 families reported worldwide** since 2011. Precise prevalence/incidence are not established (orphan-level, likely <1/1,000,000).

- **Consanguinity:** major contributor (several homozygous cases from consanguineous unions).
- **Founder effects:** none established; alleles are largely private.
- **Carrier frequency:** not formally estimated; biallelic LOF essentially absent from gnomAD.
- **Sex ratio:** no sex bias reported (autosomal). **Geographic distribution:** cases reported across multiple regions (Europe, Middle East, South Asia, East Asia), without a defined endemic focus.

10. Diagnostics

- **Genetic testing (definitive):** WES or WGS, or targeted ADAM17 sequencing, **plus** copy-number/deletion analysis and RNA/splicing assessment (needed to detect large deletions and splice variants, as in Samuelov 2022). WGS-first is advocated for faster diagnosis of such barrier/immune disorders (PMID: 39381601).
- **Functional confirmation:** cell-based shedding assays (e.g., Adam10/17 double-knockout fibroblasts) to classify missense VUS as LOF (Imoto 2021).
- **Laboratory/immunology:** impaired LPS-induced TNF-α release with elevated IL-1β/IL-6 (patient PBMCs); diminished T-cell TNF-α/IL-2 production (Bandsma 2015). These support but do not alone diagnose.
- **Histopathology:** skin biopsy shows psoriasiform/inflammatory changes with barrier abnormality; gut biopsy shows inflammatory enteropathy.
- **Differential diagnosis: NISBD2 (EGFR)** — nearly identical, distinguished only by gene; Netherton syndrome (SPINK5); other congenital ichthyosiform erythrodermas; very-early-onset/monogenic IBD (e.g., IL-10/IL-10R defects); primary immunodeficiency with skin/gut inflammation; acrodermatitis enteropathica. Genetic testing resolves the differential.
- **Screening:** no population newborn screen; **cascade carrier testing** of relatives and genetic counseling are appropriate once the familial variant is known.

11. Outcome / Prognosis

- **Mortality:** high in infancy/childhood. Documented deaths: sepsis at 10 months (Bandsma 2015); sudden death at 12 years from parvovirus B19 myocarditis (Blaydon 2011). **Cardiac disease and sepsis are the main disease-attributable causes of death.**
- **Morbidity:** chronic skin disease, malabsorption/failure to thrive, recurrent infections, hair/nail disability, esophageal stricture.

- **Recovery potential:** no cure; partial symptomatic improvement achievable (biologics for skin, budesonide for gut). Survivors face chronic lifelong disease.
- **Prognostic factors:** severity/organ extent (especially cardiac involvement and sepsis frequency); residual ADAM17 function (allele-dependent); timeliness of supportive care and infection control.

12. Treatment

No curative therapy exists; management is **supportive** + **anti-inflammatory/biologic** + **genotype-aware**.

Modality	Example	Rationale / evidence	NCIT suggestion
Anti-TNF-α biologic	infliximab/ etanercept-class	Skin improvement in combination regimen (Samuelov 2022)	NCIT:C2367 (TNF Inhibitor)
IL-12/23 blockade	ustekinumab-class	Combined with anti-TNF, "dramatically improved" skin (PMID: 34993966)	NCIT:C1972 (Monoclonal Antibody)
Enteric corticosteroid	budesonide	Controlled GI symptoms (Samuelov 2022)	NCIT:C1130 (Budesonide)
Anti-infective therapy / prophylaxis	antibiotics	Treat/prevent recurrent bacterial infection & sepsis	NCIT:C258 (Antibiotic)
Nutritional support	enteral/ parenteral nutrition	Failure to thrive, malabsorption	NCIT:C15190 (Nutritional Support)
Skin barrier care	emollients, wound care	Barrier dysfunction	—

- **Pharmacogenomics:** none disease-specific established.
- **Advanced therapeutics:** gene/cell/RNA therapies are conceptually attractive (monogenic recessive LOF) but **none are in trials** for NISBD1. Recombinant/topical EGFR-ligand supplementation (e.g., amphiregulin) is mechanistically rational and restored barrier function in ADAM17-deficient cells in vitro (PMID: 42803585) but remains experimental.

- **Personalized approach:** genotype/functional-assay–guided; biologic selection targeting the dysregulated cytokines.

13. Prevention

- **Primary prevention:** not possible (genetic). **Genetic counseling** for consanguineous/affected families; **carrier testing** and **prenatal / preimplantation genetic diagnosis** where the familial ADAM17 variant is known.
- **Secondary prevention:** early genetic diagnosis (WGS-first) to shorten the diagnostic odyssey and start supportive care/biologics sooner (PMID: 39381601).
- **Tertiary prevention:** aggressive infection prophylaxis, nutritional optimization, cardiac surveillance (given fatal myocarditis/cardiomyopathy risk), and skin-barrier management to prevent complications.
- **Immunization:** standard pediatric immunization is appropriate; no disease-specific vaccine.
- **Public-health/environmental interventions:** not applicable beyond infection control.

14. Other Species / Natural Disease

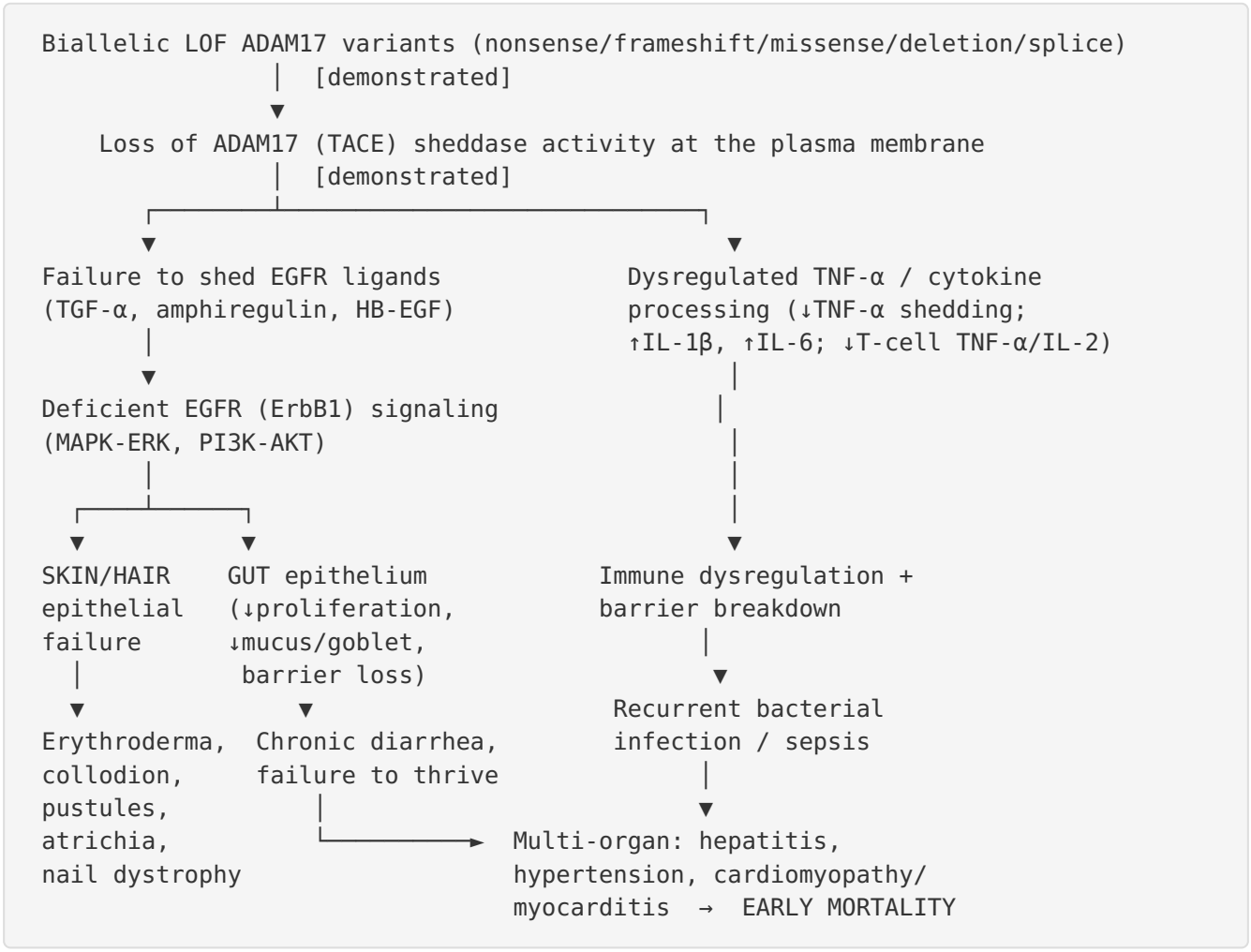
- **Taxonomy / orthologs:** Human ADAM17 (NCBI Gene 6868). Orthologs: mouse *Adam17* (NCBI Gene 11491); conserved across vertebrates (rat, zebrafish). High evolutionary conservation of the ADAM17–iRhom–EGFR axis.
- **Natural disease in other species:** No well-characterized spontaneous ADAM17-deficiency disease in companion animals is recorded in OMIA for NISBD1; knowledge is from engineered mouse models (Finding 5).
- **Comparative biology:** Adam17-null mice phenocopy EGFR/TGF-α deficiency (open-eye-at-birth, skin/epithelial defects), establishing cross-species conservation of the mechanism (Peschon 1998, PMID: 9812885).
- **Zoonotic potential:** none (genetic disease).

15. Model Organisms

Model	Type	Genetic strategy	Recapitulation	Key limitation	PMID
Adam17-null mouse	Mammalian, in vivo	Germline knockout	Epithelial/skin defects, phenocopy of EGFR/TGF-α loss	Perinatal/ embryonic lethality limits adult study	9812885
Tace/Sox9 conditional mouse	Mammalian, in vivo	Hair-follicle-SC conditional KO	Brittle hair, progressive alopecia, bulge-cell loss	Tissue-restricted; models hair arm only	22696231
Intestinal/ systemic Adam17 KO mouse	Mammalian, in vivo	Conditional/ systemic KO	Severe DSS colitis, impaired EGFR-dependent epithelial protection	Chemically induced colitis, not spontaneous	27077118
iRhom cub/ toc mice	Mammalian, in vivo	iRhom2 gain/ loss	ADAM17–amphiregulin–EGFR-dependent skin/hair phenotypes	Models cofactor axis, not ADAM17 LOF directly	37282854
Adam10/17 double-KO MEFs	In vitro cellular	Double knockout fibroblasts	Shedding-assay platform to test human variants	No tissue architecture	33953303

These models collectively reproduce the skin-barrier, hair-follicle, and bowel-inflammation arms of NISBD1 and causally tie them to impaired EGFR-ligand shedding. **Resource databases:** MGI, IMPC, IMSR for the mouse alleles; Cellosaurus for the Adam10/17-DKO fibroblast line.

Mechanistic Model / Interpretation



The decisive evidence that this chain is correct — rather than merely plausible — is the **NISBD1 (ADAM17-loss) ↔ NISBD2 (EGFR-loss) allelic series** (Finding 8): knocking out either the enzyme that makes the ligand or the receptor it activates produces near-identical neonatal skin + bowel + hair disease, and both are mirrored by the opposite (gain-of-function iRHOM2) disease TOC. Human genetics, mouse models, and structural biology thus triangulate on one axis.

Evidence Base

PMID	Study	Role in this report
22010916	Blaydon 2011 (NEJM)	Founding paper: ADAM17 LOF causes NISBD1; cytokine data; cardiac mortality
25804906	Bandsma 2015	Second family (c.308dupA); multi-organ dysfunction; sepsis death
33953303	Imoto 2021	Compound-het missense LOF, functionally confirmed; ADAM17 >80 substrates
34993966	Samuelov 2022	Complex allele; anti-TNF + IL-12/23 + budesonide response
40968583	Chang 2026	Core phenotype spectrum definition
42015567	Malhi 2026	Atypical phenotype; expressivity; novel allele (p.Gln146Lys)
9812885	Peschon 1998 (Science)	TACE-null mouse; ectodomain shedding essential; EGFR phenocopy
18762209	Edwards 2008	ADAM17 generates active EGFR ligands; epithelial development
22696231	Nagao 2012	Hair-follicle phenotype model
27077118	Shimoda 2016	Colitis/EGFR epithelial protection model
37282854	Rabinowitsch 2023	iRhom–ADAM17–amphiregulin–EGFR dependence
38781971	Lu 2024 (cryo-EM)	iRhom2 controls ADAM17 maturation/activity
40512800	Maciag 2025 (cryo-EM)	ADAM17–iRhom2 activation/inhibition structure
24643277	Brooke 2014	GOF iRHOM2/ADAM17 in TOC — mirror-image skin disease
40040597	Karabinos 2025	Defines NISBD2 (EGFR) counterpart
9562519	Miettinen 1997	EGFR-null epithelial/intestinal disease
39381601	2024	WGS-first diagnosis of barrier/immune disorders

Evidence-type mix: human clinical case reports/series (diagnosis, phenotype, treatment); in vitro functional assays (variant validation); mouse models (causal mechanism); cryo-EM structural biology (cofactor regulation). This multi-modal convergence makes the ADAM17→EGFR-ligand→EGFR model robust despite the tiny patient population.

Limitations and Knowledge Gaps

1. **Extremely small sample:** <10 families worldwide; no cohort-level prevalence, incidence, penetrance, or natural-history statistics. All frequencies are qualitative.
2. **Phenotype–genotype correlation unresolved:** why some patients lack bowel disease, or develop cardiomyopathy/hepatitis, is unexplained; modifier genes (iRhom paralogs, EGFR-axis genes) are hypothesized but unproven in humans.
3. **Mechanism of extracutaneous/extraintestinal organ damage** (cardiac, hepatic, vascular) is observed but not mechanistically dissected; whether it is primary (ADAM17 loss in those tissues) or secondary (systemic inflammation/infection) is unclear.
4. **Treatment evidence is anecdotal:** the anti-TNF + IL-12/23 regimen rests on single-patient experience, not controlled trials; optimal regimen and durability are unknown.
5. **No curative or disease-modifying therapy**; gene/EGFR-ligand-replacement approaches are untested in patients.
6. **No validated QoL or outcome instruments** for this ultra-rare condition.

Proposed Follow-up Experiments / Actions

1. **International patient registry & genotype–phenotype database** — pool the <10 known families to quantify penetrance, organ-involvement frequencies, mortality, and allele-specific severity.
2. **Standardized functional-assay pipeline** for all new ADAM17 variants (Adam10/17-DKO fibroblast shedding) to classify VUS and correlate residual activity with severity.
3. **Mechanistic study of cardiac/hepatic involvement** using tissue-specific Adam17 conditional mice to determine whether organ damage is cell-autonomous or inflammation-driven.

4. **Preclinical EGFR-ligand/pathway rescue** — test topical/recombinant amphiregulin or EGFR agonism for skin/gut barrier restoration (rationale: AREG rescued barrier in ADAM17-deficient epithelium in vitro, PMID: 42803585).
5. **Prospective evaluation of biologic regimens** (anti-TNF ± IL-12/23, and alternatives) via an N-of-1/registry-embedded framework given rarity.
6. **Cascade carrier screening and reproductive counseling** (PGD/prenatal testing) offered systematically to consanguineous and affected families.
7. **Early-WGS diagnostic pathway** adoption to shorten the diagnostic odyssey and enable earlier supportive/biologic intervention.

Report compiled from 8 confirmed findings across 33 reviewed papers. All quoted abstract text is verbatim from the cited PMIDs. Evidence types are distinguished throughout as human clinical, model organism, in vitro, or structural/computational.
